

Roll No. :

MID SEMESTER EXAMINATION 2025

Name of the Program: **B.Tech (CSE)**

Semester: 5TH

Name of the Course: Computer Network-I

Course Code: TCS 514

Time : 90 Minutes

Maximum Marks : 50

Note:

- i. Answer all the questions by choosing any one of the sub questions
- ii. Each questions carries 10 marks

Q1		(10 marks)	CO-1
a)	What is the difference between a host and an end system? List several different types of end systems. Is a Web server an end system?		
OR			
b)	What advantage does a circuit-switched network have over a packet-switched network? What advantages does TDM have over FDM in a circuit-switched network?		CO-1
Q2			
(10 marks)			
a)	What are the three phases of circuit switching? List the advantages and disadvantages of circuit switching and packet switching? What are the delays in packet switching?		CO-1
OR			
b)	Consider sending a packet from a source host to a destination host over a fixed route. List the delay components in the end-to-end delay. Which of these delays are constant and which are variable		
Q3		(10 marks)	CO-1, CO2, CO3
a)	Let's consider a simple scenario where a packet travels from Host A to Host B through one router. If Packet size is 1500 bytes (12000 bits), data Transmission Rate is 10 Mbps (10,000,000 bits per second), Link Distance from Host A to Host B is 1000 km, data propagation speed is 2×10^8 meters per second, Queuing Delay is 2 ms, Processing Delay is 1 ms. Assuming the router adds no further delay, then calculate end-to-end delay of packet traversal from Host A to Host B.		
OR			
b)	What is the HOL blocking issue in HTTP/1.1? How does HTTP/2 attempt to solve it?		CO-2, CO3
Q4			
(10 marks)			
a)	Describe how Web caching can reduce the delay in receiving a requested object. Will Web caching reduce the delay for all objects requested by a user or for only some of the objects? Why?		CO-2, CO3
OR			
b)	Discuss peer-to peer architecture and for a P2P file-sharing application, do you agree with the statement, "There is no notion of client and server sides of a communication session"? Why or why not?		
Q5		(10 marks)	CO-3,
CO3	Why do HTTP, SMTP, and IMAP run on top of TCP rather than on UDP?		
OR			
b)	Why is it that voice and video traffic is often sent over TCP rather than UDP in today's Internet?		CO-3

